

Mine-resistant ambush protected vehicle

Mine-resistant ambush protected vehicle (**MRAPV**), also known as **MRAP vehicle**, is a type of armoured personnel carrier that is designed specifically to withstand the impact of land mines, improvised explosive device (IED) attacks, and ambushes.^{[1][2][3][4]} Most modern infantry mobility vehicles also have a certain level of MRAP capabilities.

History

Specialized light armored vehicles designed specifically to resist land mines were first introduced in the 1970s by the Rhodesian Army, and were further developed by South African manufacturers starting in 1974 with the Hippo armored personnel carrier (APC).^{[5][6]} The first step by the South African Defence Force (SADF) was the Bosvark, a Unimog fitted with a shallow mine-deflecting tub on the chassis to protect the crew. Then came the first generation of purpose-built vehicles, including the Hippo and various other light vehicles. They were essentially armoured V-shaped hulls mounted on truck chassis.

The next generation was represented by the Buffel, a Unimog chassis with a mine-protected cab and a mine-protected crew compartment mounted on it. These early vehicles overloaded their chassis and they were clumsy off-road. The Casspir Mine-Resistant Ambush Protected Vehicle was developed for the SADF after 1980;^[7] this was the inspiration for the American and other military MRAPV program and the basis for some of the program's vehicles.^{[8][9][10]}

Design

These vehicles have good off-road mobility, and armor protection against small-arms fire, IEDs, and antipersonnel mines. These armored vehicles generally have a distinctive V-shaped hull (for mine protection) and a wheeled chassis.^{[1][2][3][4]}



United States-made Cougar HE MRAPV being tested in January 2007 with landmines



Turkish-made BMC Kirpi in Ukrainian Marine Corps service

List of MRAP vehicles

Dedicated MRAPV			
Name	Country of origin	Entered service	Image
<u>Hippo</u>	<u>South Africa</u>	1974	
<u>Buffel</u>	<u>South Africa</u>	1978	
<u>MPCV "Spook"</u>	<u>Rhodesia</u>	1979	

<u>Mamba</u>	<u>South Africa</u>	1990	
<u>ATF Dingo</u>	<u>Germany</u>	2000	
<u>Unibuffel</u>	<u>Sri Lanka</u>	2000	
<u>Cougar</u>	<u>United States</u>	2002	
<u>RG-31 Nyala</u>	<u>South Africa</u>	2006	

<u>BAE Caiman</u>	<u>United States</u>	2007	
<u>Gaz Tigr</u>	<u>Russia</u>	2007	
<u>International MaxxPro</u>	<u>United States</u>	2007	
<u>RG-33</u>	<u>South Africa</u>	2007	
<u>Marauder</u>	<u>South Africa</u>	2008	

<u>BMC Vuran</u>	<u>Turkey</u>	2009	
<u>Nexter Aravis</u>	<u>France</u>	2009	
<u>Oshkosh M-ATV</u>	<u>United States</u>	2009	
<u>First Win</u>	<u>Thailand</u>	2010	
<u>Force Protection Ocelot</u>	<u>United Kingdom</u>	2011	

<u>CS/VP3 MRAP</u>	<u>China</u>	2012	
<u>BMC Kirpi</u>	<u>Turkey</u>	2014	
<u>Kamaz Typhoon</u>	<u>Russia</u>	2014	
<u>Protolab Misu</u>	<u>Finland</u>	2017	
<u>Sisu GTP</u>	<u>Finland</u>	2018	

<u>Toofan</u>	<u>Iran</u>	2018	
<u>Ezugwu MRAP</u>	<u>Nigeria</u>	2019	
<u>JLTV</u>	<u>United States</u>	2019	
<u>Unicob</u>	<u>Sri Lanka</u>	2021	
<u>Kalyani M4</u>	<u>India</u>	2022	

<u>Zastava M20 MRAP</u>	<u>Serbia</u>	2023	
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Infantry mobility vehicle with MRAP capabilities

- AM
- M16 Miloš
- Bushmaster IMV
- COMBATGUARD
- Didgori series
- Grizzly APC
- Hunter TR-12
- Iveco LMV – several thousand ordered by Italian and other European militaries
- Kozak (armored personnel carrier)
- Lenco BearCat G5
- Nurol Ejder
- Mahindra Armored Light Specialist Vehicle
- Mungo ESK
- Oshkosh L-ATV – selected to meet US military's JLTV requirement on 25 August 2015
- Otokar Cobra
- Otokar Cobra II
- RG-33 (4×4)
- Shota
- STREIT Group Spartan
- Varta

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